

# Spectrum reference cards (what each source looks like)

Look at each light through your diffraction glasses and match what you see to a card. The key idea: two lights can both look white but have very different spectra. Sketches are schematic — match the PATTERN (smooth rainbow vs broad band vs separate lines), not the exact colors.

## Filament / incandescent bulb



A smooth, complete rainbow with no gaps. A hot solid glows across all colors (a continuous spectrum).

## White LED flashlight



A bright blue spike plus a broad band of green-yellow-orange, weaker deep red. A blue LED chip plus a yellow phosphor, NOT separate lines.

## Neon / gas light



A black background with a few separate bright lines (neon is orange-red). Each gas has its own line fingerprint — this is how we read starlight and fireworks.

**Your task:** sketch the spectrum of each mystery source, then name it (filament, LED, or neon/gas) and give one reason from its pattern. A line spectrum is a fingerprint; a broad band is not.